

CRM Project Documentation

1. Project Overview

This CRM system is built to simplify and organize the complete sales process in one centralized platform. The application helps sales teams manage leads, customers, deals, meetings, calls, and follow-ups without depending on manual tracking methods.

The system focuses on improving productivity by combining human actions with smart automation. Instead of replacing user control, the CRM assists users by automatically handling repetitive operations like task creation, status updates, and workflow transitions.

Main Purpose of the System

- Manage the complete customer lifecycle from lead generation to deal closure.
- Improve follow-up management and sales tracking.
- Reduce manual work using workflow automation.
- Provide better visibility into sales performance and pipeline movement.

Main Users

- **Sales Representatives** — Handle leads, calls, meetings, and deal progress.
 - **Managers** — Monitor team performance and pipeline activities.
 - **Administrators** — Manage users, permissions, workflows, and system configurations.
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2. Problem Statement

Before implementing this CRM, most sales activities were managed manually using spreadsheets, calls, notes, or disconnected tools. This created several operational problems.

Existing Problems

- Sales teams struggled to track customer interactions properly.
- Follow-ups were often missed due to lack of reminders.
- Deal stages were updated manually, causing inconsistencies.
- Teams spent excessive time creating repetitive tasks.
- Pipeline visibility was poor for managers.
- Communication history was scattered across different places.

Business Impact

- Reduced conversion rates.
- Delayed follow-ups.
- Inaccurate sales tracking.

- Productivity loss due to manual operations.
 - Lack of real-time visibility into sales progress.
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3. Proposed Solution

The CRM application solves these challenges by introducing a centralized and automation-driven workflow management system.

Solution Approach

- All customer data is stored in one unified platform.
- Automated workflows reduce repetitive manual operations.
- Task chaining automatically creates the next action based on previous outcomes.
- Real-time synchronization keeps all users updated instantly.
- Human-driven automation ensures users still maintain full control over decisions.

Key Improvements

- Faster lead follow-up process.
 - Better pipeline visibility.
 - Reduced manual data entry.
 - Improved workflow consistency.
 - Centralized communication and activity tracking.
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4. Objectives

Main Project Goals

- Build a scalable and reliable CRM platform.
- Simplify the sales lifecycle process.
- Improve customer relationship management.

Automation Goals

- Automatically create follow-up tasks.
- Reduce repetitive user operations.
- Trigger workflows based on user activities.

Productivity Goals

- Allow sales teams to focus more on selling.
- Reduce administrative workload.
- Improve task management efficiency.

CRM Improvement Goals

- Increase lead conversion tracking.
 - Improve data consistency.
 - Enhance reporting and pipeline monitoring.
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5. Scope of the Project

The CRM system covers the entire sales management process from lead handling to deal closure.

Features Included

- Lead management
- Contact management
- Account management
- Deal pipeline tracking
- Task management
- Call logging
- Meeting scheduling
- Workflow automation
- Notification system
- Reports and analytics
- User and role management

Modules Included

- Authentication
- Dashboard
- Leads
- Contacts
- Accounts
- Deals/Pipeline
- Tasks
- Calls
- Meetings
- Workflow Engine
- Notifications
- Reports
- Settings

System Boundaries

The current system mainly focuses on internal CRM operations and sales workflow automation. External communication integrations and advanced forecasting are limited in the current implementation.

6. Functional Requirements

Authentication

- Secure login and logout functionality.
- JWT/token-based authentication system.
- Automatic token refresh mechanism.

User Management

- Role-based access control.
- User assignment and team management.
- Admin-level configuration access.

Lead Management

- Create, update, and manage leads.
- Track lead statuses.
- Maintain activity history.
- Convert leads into deals and contacts.

Task Management

- Support multiple task types:
 - Calls
 - Meetings
 - Proposals
 - Emails
 - To-Do tasks
- Track task progress and outcomes.
- Trigger workflows after task completion.

Contact & Account Management

- Store customer contact information.
- Link contacts with accounts/companies.
- Maintain relationship mapping.

Deal/Pipeline Management

- Track deals across pipeline stages.
- Manage deal probabilities.
- Handle deal closure status.
- Prevent duplicate active deals.

Call & Meeting Management

- Schedule and log calls.
- Live call timer support.
- Manage meeting statuses and durations.

Workflow Automation

- Trigger actions automatically based on events.
 - Support conditional logic.
 - Execute actions such as:
 - Task creation
 - Status updates
 - Notifications
 - Owner assignment
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7. Project Modules

Authentication Module

Purpose

Provides secure access to the CRM system.

Features

- Login
- Logout
- Registration
- Token refresh

Database

Uses Django user authentication with extended role support.

Dashboard Module

Purpose

Provides an overview of sales activities and performance.

Features

- Lead summaries
 - Deal summaries
 - Task statistics
 - Performance charts
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Leads Module

Purpose

Manage potential customers.

Features

- Lead creation
- Lead updates
- Status tracking
- Activity timeline

Database

Lead model connected with contacts and tasks.

Deals/Pipeline Module

Purpose

Manage sales opportunities and deal movement.

Features

- Pipeline stages
- Deal tracking
- Probability calculations
- Won/Lost management

Database

Deal model linked with leads and accounts.

Tasks Module

Purpose

Handle daily sales activities.

Features

- Call tasks
- Meeting tasks
- Proposal tasks
- Outcome-driven workflows

Database

Task model connected to leads, contacts, accounts, and deals.

Workflow Engine Module

Purpose

Automate business processes.

Features

- Trigger-based automation
- Rule execution
- Conditional workflows
- Debouncing and validation checks

Database

Workflow and WorkflowRule models.

8. Workflow Process

Lead Creation Flow

1. A new lead is created.
2. System automatically assigns a follow-up task.
3. Sales representative contacts the lead.

Call Process

1. User starts a call task.

2. Call timer tracks duration.
3. Outcome is selected after completion.
4. Workflow triggers based on outcome.

Meeting Process

1. Meeting task gets scheduled.
2. Meeting completion updates lead progress.
3. Follow-up tasks may be created automatically.

Lead Qualification Flow

1. Lead status changes after successful interactions.
2. Qualified leads proceed to deal creation.

Deal Creation Flow

1. Lead converts into:
 - Contact
 - Account
 - Deal
2. Deal enters pipeline stages.

Pipeline Movement

- Deals move through sales stages based on task outcomes and user updates.

Deal Closure

- Deal marked as:
 - Won
 - Lost
 - Final workflow actions execute automatically.
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9. Technology Stack

Frontend Technologies

- React
- Vite
- TailwindCSS
- Axios
- Lucide React
- Dayjs

Backend Technologies

- Django
- Django REST Framework

Database

- SQLite (Current development environment)

Real-Time Communication

- WebSockets
- Django Channels

Background Processing

- Celery-based task handling (implied from architecture)
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10. Database Design

Main Entities

User

Stores user information, roles, and teams.

Lead

Stores prospect details and lead status information.

Deal

Stores sales opportunity details.

Task

Stores activities like calls, meetings, proposals, and follow-ups.

Contact

Stores customer contact information.

Account

Stores company/account details.

Workflow

Stores automation configurations.

Relationships

- Lead → Contact
 - Deal → Lead
 - Deal → Account
 - Task → Lead
 - Task → Deal
 - Task → Contact
 - Contact → Account
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11. Additional Analysis

Existing Automations

- Automatic task creation after lead generation.
- Workflow execution after task completion.
- Lead conversion automation.
- Deal stage progression workflows.

Existing Validations

- Mandatory close date for deals.
- Positive deal value validation.
- Single active deal restriction per lead.

Current Limitations

- SQLite is not ideal for large-scale production usage.
- Duplicate models may create synchronization complexity.
- Limited external integrations.

Missing Features

- Advanced analytics dashboards.
- Forecasting modules.
- Email/VOIP integrations.

- AI-based recommendations.

Scalability Observations

The architecture supports future scaling through modular design, workflow abstraction, and real-time communication support.